Circles

Lesson 1

Teacher's copy: overview.

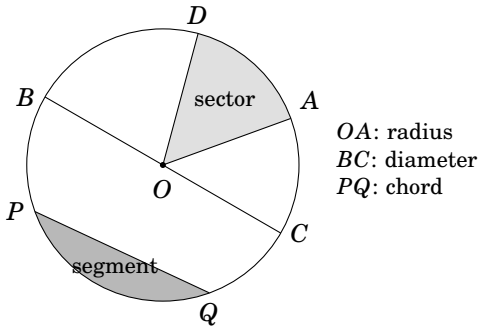
- **Aims.** Students use the terms of a circle correctly. They explain why equal chords subtend equal angles at the centre, and the converse. They explain why the perpendicular from the centre bisects a chord, and the converse. They find chord lengths and distances with the Pythagoras theorem.
- **Syllabus.** CBSE Class IX 2026–27, Unit IV Geometry (25 of 80 marks). The syllabus asks students to *explain why* each result holds. The explanations here use congruence (SSS, SAS, RHS).
- **Materials.** Compass, ruler, set square, protractor, paper circles cut beforehand, scissors.
- **Timing.** Section 1: 5 min. Section 2: 15 min. Section 3: 20 min. Practice: 15 min. Exit question: 5 min.
- **Next lesson.** The circle through three points and the circumcircle of a triangle.

1 Parts of a circle (5 min)

Start with familiar circles: the wheel of a bullock cart and the Ashoka Chakra on the national flag. Ask: what is the same about every spoke of a wheel?

Definition 1. A **circle** is the set of all points in a plane at a fixed distance from a fixed point. The fixed point is the **centre**. The fixed distance is the **radius**.

- A **chord** is a line segment that joins two points on the circle.
- A **diameter** is a chord through the centre. Diameter = 2× radius.
- An **arc** is a part of the circle. A chord cuts the circle into a **minor arc** and a **major arc**.
- A **segment** is the region between a chord and an arc.
- A **sector** is the region between two radii and an arc.

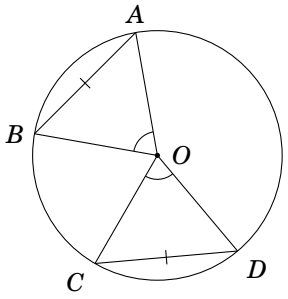


Note for the teacher. Ask students to draw a circle and name each part on their own drawing. Students often confuse a segment with a sector. A sector always has two radii as its sides; a segment has a chord.

2 Equal chords and angles at the centre (15 min)

Activity. Draw a circle with centre O. Draw a chord AB. Use the compass to copy the length AB and draw another chord CD of the same length. Join OA, OB, OC and OD. Measure $\angle AOB$ and $\angle COD$. What do you notice?

Theorem 1. *Equal chords of a circle subtend equal angles at the centre.*

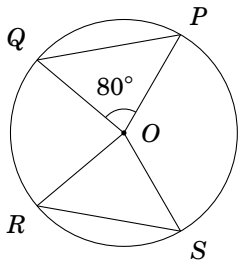


Why it works. In $\triangle AOB$ and $\triangle COD$: $OA = OC$ (radii), $OB = OD$ (radii) and $AB = CD$ (given). So $\triangle AOB \cong \triangle COD$ (SSS). Hence $\angle AOB = \angle COD$ (corresponding parts of congruent triangles).

Theorem 2 (Converse)**.** *If two chords of a circle subtend equal angles at the centre, then the chords are equal.*

Why it works. In $\triangle AOB$ and $\triangle COD$: $OA = OC$, $\angle AOB = \angle COD$ and $OB = OD$. So $\triangle AOB \cong \triangle COD$ (SAS). Hence $AB = CD$.

Example 2.1. In a circle with centre O, chords PQ and RS are equal and $\angle POQ = 80°$. Find $\angle ROS$ and $\angle ORS$.



Given: $PQ = RS$ and $\angle POQ = 80°$.
To find: $\angle ROS$ and $\angle ORS$.

1. $\angle ROS = \angle POQ = 80°$ (equal chords subtend equal angles at the centre, Theorem 1).
2. $OR = OS$ (radii), so $\triangle ORS$ is isosceles and $\angle ORS = \angle OSR$.
3. $\angle ORS + \angle OSR = 180° - 80° = 100°$ (angle sum of a triangle), so $\angle ORS = 50°$.

Answer: $\angle ROS = 80°$ and $\angle ORS = 50°$.

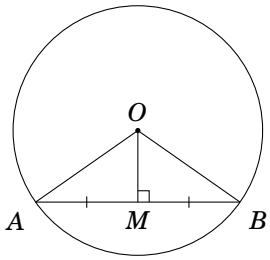
Note for the teacher. Stress that the angles are at the *centre*. Ask: "Would the result hold if the angles were measured at a point on the circle?" Leave the question open for a later lesson.

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3 The perpendicular from the centre to a chord (20 min)

Activity. Cut out a paper circle and mark its centre O. Fold the paper once to make a crease AB; this is a chord. Now fold again so that A falls exactly on B. Open the paper. Where does the new crease cross AB? Does it pass through O? Measure the angle between the two creases.

Theorem 3. *The perpendicular from the centre of a circle to a chord bisects the chord.*



Why it works. Let $OM \perp AB$. In $\triangle OMA$ and $\triangle OMB$: $\angle OMA = \angle OMB = 90°$, $OA = OB$ (radii) and OM is common. So $\triangle OMA \cong \triangle OMB$ (RHS). Hence $AM = BM$.

Theorem 4 (Converse)**.** *The line joining the centre of a circle to the midpoint of a chord is perpendicular to the chord.*

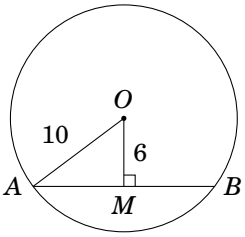
Why it works. Let M be the midpoint of AB. In $\triangle OMA$ and $\triangle OMB$: $AM = BM$, $OA = OB$ (radii) and OM is common. So $\triangle OMA \cong \triangle OMB$ (SSS) and $\angle OMA = \angle OMB$. These two angles form a linear pair, so each is $90°$.

Things to Remember

The **distance of a chord from the centre** is the length of the perpendicular from the centre to the chord. If the radius is r, the chord has length c and its distance from the centre is d, then the right triangle OMA gives

$$\left(\frac{c}{2}\right)^2 + d^2 = r^2.$$

Example 3.1. A chord of a circle of radius 10 cm is 6 cm from the centre. Find the length of the chord.



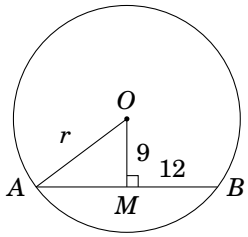
Given: radius $OA = 10$ cm, $OM \perp AB$, $OM = 6$ cm.
To find: the length of chord AB.

1. $AM = MB$ (the perpendicular from the centre bisects the chord, Theorem 3).

2. $\triangle OMA$ is right-angled at M, so $AM^2 = OA^2 - OM^2 = 100 - 36 = 64$ (Pythagoras theorem). So $AM = 8$ cm.
3. $AB = 2 \times AM = 2 \times 8 = 16$ cm.

Answer: the chord is 16 cm long.

Example 3.2. A chord of length 24 cm is 9 cm from the centre of a circle. Find the radius of the circle.



Given: $AB = 24$ cm, $OM \perp AB$, $OM = 9$ cm.
To find: the radius OA.

1. $AM = \dfrac{24}{2} = 12$ cm (Theorem 3).
2. $OA^2 = AM^2 + OM^2 = 144 + 81 = 225$ (Pythagoras theorem in $\triangle OMA$). So $OA = 15$ cm.

Answer: the radius is 15 cm.

Note for the teacher. Students often use the whole chord in place of half of it. Ask them to mark the midpoint M and the right angle before they calculate.

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4 Practice (15 min)

Work in your exercise book. Draw a rough diagram for every question.

Exercise 1A

1. The radius of a circle is 13 cm. A chord is 5 cm from the centre. Find the length of the chord. Draw a rough diagram first.
2. A chord of length 30 cm is drawn in a circle of radius 17 cm. Find the distance of the chord from the centre.
3. Chord AB of a circle with centre O is 48 cm long. $OM \perp AB$ and $OM = 7$ cm. Ravi writes: radius $= \sqrt{48^2 + 7^2}$. Find his mistake, and find the correct radius.
4. In a circle with centre O, chords AB and CD are equal and $\angle AOB = 70^\circ$. Find $\angle COD$ and $\angle OCD$. Give a reason for each step.
5. Draw a circle of radius 5 cm and a chord of length 8 cm in it. Use a set square to draw the perpendicular from the centre to the chord. Measure its length. Then find the length by calculation and compare.

6. A round dining table top has radius 50 cm. A carpenter fixes a straight wooden strip on the underside along a chord, 30 cm from the centre. Find the length of the strip.

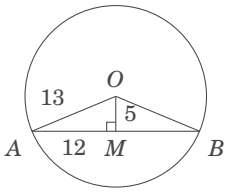
Note for the teacher. If time is short, set Questions 5 and 6 as homework. Questions 1 to 3 give the main practice in the relation $\left(\frac{c}{2}\right)^2 + d^2 = r^2$.

Answers.

1. $OM \perp AB$, so M is the midpoint of AB (Theorem 3). In right triangle OMA,

$$AM^2 = OA^2 - OM^2 = 13^2 - 5^2 = 169 - 25 = 144,$$

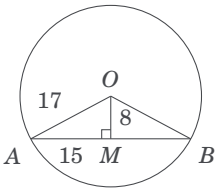
so $AM = 12$ cm, and $AB = 2 \times 12 = 24$ cm. **Answer:** 24 cm.



2. $AM = \frac{30}{2} = 15$ cm. In right triangle OMA,

$$OM^2 = OA^2 - AM^2 = 17^2 - 15^2 = 289 - 225 = 64,$$

so $OM = 8$ cm. **Answer:** 8 cm.



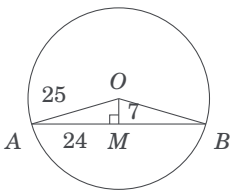
3. Ravi used the whole chord in place of the half chord. M is the midpoint of AB, so

$$AM = \frac{48}{2} = 24 \text{ cm}.$$

In right triangle OMA,

$$OA^2 = AM^2 + OM^2 = 24^2 + 7^2 = 576 + 49 = 625,$$

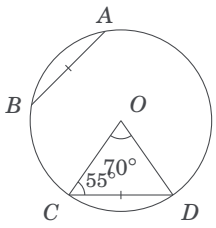
so $OA = 25$ cm. **Answer:** the radius is 25 cm.



4. Equal chords subtend equal angles at the centre, so $\angle COD = \angle AOB = 70^\circ$ (Theorem 1). In triangle OCD, $OC = OD$ because both are radii, so the triangle is isosceles and $\angle OCD = \angle ODC$. Therefore

$$\angle OCD = \frac{180^\circ - 70^\circ}{2} = \frac{110^\circ}{2} = 55^\circ.$$

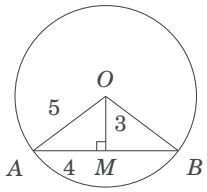
Answer: $\angle COD = 70^\circ$, $\angle OCD = 55^\circ$.



5. The half chord is $AM = 4$ cm. In right triangle OMA,

$$OM^2 = OA^2 - AM^2 = 5^2 - 4^2 = 25 - 16 = 9,$$

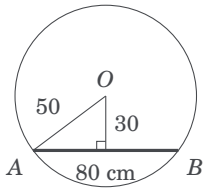
so $OM = 3$ cm. The measured length should be close to 3 cm. **Answer:** 3 cm.



6. The strip is a chord. The half chord is

$$\sqrt{50^2 - 30^2} = \sqrt{2500 - 900} = \sqrt{1600} = 40 \text{ cm},$$

so the strip is $2 \times 40 = 80$ cm long. **Answer:** 80 cm.



5 Exit question (5 min)

Answer on a slip of paper.

(a) True or false? "Every line through the midpoint of a chord passes through the centre." Give a reason.

(b) A chord of length 10 cm is drawn in a circle of radius 13 cm. How far is the chord from the centre?

Answers.

(a) False. The line joining the centre to the midpoint of a chord is perpendicular to the chord (Theorem 4). A slanted line through the midpoint does not pass through the centre. Only the perpendicular from the centre bisects the chord (Theorem 3). **Answer:** false.

(b) The half chord is 5 cm. In right triangle OMA,

$$d^2 = 13^2 - 5^2 = 169 - 25 = 144,$$

so $d = 12$ cm. **Answer:** 12 cm.

